MILITARY SPECIFICATION

FLARE, SURFACE, PARACHUTE, XM183 PARTS FOR AND LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the parts and loading, assembling and packing for one type of Flare,Surface, Parachute designated as XM183.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-223	- Powder, Black
MIL-A-2550	- Ammunition and Special Weapons; General Specification for
MIL-W-6858	- Welding, Resistance: Aluminum, Magnesium, Nonhardening Steels or Alloys, Heat-Resisting Alloys, and Titanium Alloys; Spot and Seam

STANDARDS

FEDERAL

FED-STD-406	- Plastics; Method of Testing

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	- Quality Assurance Terms and Definitions

FSC: 1370

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MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1234 - Pyrotechnics; Sampling, Inspection and Testing
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U. S. ARMY MUNITIONS COMMAND

9229070 - Flare, Surface, Parachute, XM183
9241109 - Support, Packing, Ammo, For Flare, Surface, Parachute, XM183
9235794 - Box, Wirebound, Packing, Ammunition, for Flare, Surface, Parachute, XM183

PUBLICATIONS

U. S. ARMY MUNITIONS COMMAND

ET 9229070 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assembly.-The assembly shall comply with all requirements specified on Drawing (dwg.) 9229070 and with all requirements specified in applicable specifications.

3.3 Moisture content

3.3.1 Paper and paper products.-The moisture content of the paper and paper products of the flare assembly at the loading station at the time of loading shall not exceed 0.3 percent when tested as specified in 4.4.1.1.

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3.3.2 First fire pellet.-The moisture content of the first fire pellet at the loading station at the time of loading shall not exceed 0.3 percent when tested as specified in 4.4.1.3.

3.3.3 Flare composition.-The moisture content of the flare composition at the loading station at the time of loading shall not exceed 0.3 percent when tested as specified in 4.4.1.3.

3.4 Static functioning of candle assembly.

3.4.1 Candlepower.-The individual candle assembly shall exhibit a minimum (min.) average output of 152,500 candles for the min. burning time when tested as specified in 4.4.4.

3.4.2 Burning time.-The burning time of the individual candle assembly shall be not less than 37 seconds nor more than 45 seconds when tested as specified in 4.4.4.

3.5 Leakage of projectile package assembly.-The sealed projectile package assembly shall show no evidence of leakage when subjected to the test specified in 4.4.8.

3.6 Transportation-vibration.-The launcher platform assembly and the projectile package assembly shall comply with the following requirements after being subjected to the transportation vibration test specified in Standard MIL-STD-331, Test 104, Test Procedure II. The test shall be performed as specified in 4.4.7.

3.6.1 Launcher platform assembly.-The launcher platform assembly shall show no evidence of damage or disassembly that would affect normal functioning.

3.6.2 Projectile package assembly.-The projectile package assembly shall meet the functioning requirements specified in 3.7.

3.7 Functioning.-The projectile package assembly shall function and comply with the following requirements when tested as specified in 4.4.9.

3.7.1 Premature.-The projectile assembly shall not burst in or within 100 feet of the projectile package assembly or eject in less than five (5) seconds after actuation of the fuze starter.

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3.7.2 Ignition.-The projectile package assembly ignition train shall ignite upon actuation of the fuse starter.

3.7.3 Ejection.-Each projectile assembly shall eject from the projectile package assembly.

3.7.4 Range.-The range of the projectile assembly shall be 625 ± 55 meters from point to launch.

3.7.5 Deployment altitude.-The candle assembly shall deploy at an altitude of not less than 390 or more than 780 feet above the elevation of the launcher.

3.7.6 Parachute opening.-The parachute shall open completely without evidence of partial opening, tangled, twisted or broken shroud lines or separation from the candle assembly.

3.7.7 Burning time.-The total burning time of each group of six (6) candles suspended following launching from the projectile package assembly shall be not less than two and one-half minutes. The burning time shall be measured from ignition of the first candle ignition in the group to that point at which no light is visible from the last candle in the group to ignite.

3.8 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, crazes, porosity, warpage, burn marks, checks, blisters, excess flash, unblended radii, chipped edges, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined and all packing components shall be dry.

3.9 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use

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his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the project being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing and specification number and date, together with identification and date of changes thereto.
e. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents. These statements of conformance shall be on file and available for review. Statements of findings on all compositions shall be submitted with each lot.
f. Number of items in the lot.
g. Date submitted.
h. A statement that the components and assemblies were dried and assembled as required by the drawing.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.

4.2.1 Submission.-The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of the

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following quantities which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract:

a. Twenty-five (25) complete sets of inert parts (i.e. 25 each of every inert component and inert sub-assembly).
b. Twenty (20) candle assemblies.
c. Five (5) flare assemblies completely packaged and sealed as required by the applicable drawings. All parts and materials, including packaging and packing components, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and he shall submit a record of this inspection with the sample, including statements of findings and certificates of conformance of materials. A sample containing known defects shall not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government. In the event or rejection, the Government reserves the right to require the contractor to take corrective action and submit an additional sample or portion thereof of items for first article inspection. Unless otherwise directed by the contracting officer, loaded components shall not be submitted for proving ground testing until all other phases of the first article sample have been accepted. Until that phase of the first article sample consisting of inert parts and loaded items is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.2.2 First article inspection provisions.

4.2.2.1 Inert parts

4.2.2.1.1 The 25 sets of inert parts (see 4.2.1a) shall be submitted to a Government approved laboratory designated by the contracting officer.

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4.2.2.1.2 Inspections to be performed.-The inert parts will be subjected to any or all of the applicable examinations and tests specified in 4.3.2 and 4.3.3 of this specification and any or all of the requirements of the applicable drawings.

4.2.2.1.3 Rejection.-If any inert component or sub-assembly fails to comply with any or all of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an inert component or sub-assembly to comply with any of the stated requirements.

4.2.2.2 Candle assemblies.

4.2.2.2.1 Submission.-The twenty (20) candle assemblies (see 4.2.1 b) shall be submitted for candlepower and burning time tests to Picatinny Arsenal, Dover, New Jersey.

4.2.2.2.2 Inspection to be performed.-The candle assemblies will be subjected to the static functioning test of candlepower and burning time (see 3.4) in accordance with the method specified in 4.4.4.

4.2.2.2.3 Rejection.-The first article sample shall be rejected if any candle assembly fails to comply with the requirements specified. The Government reserves the right to terminate its inspection upon encountering any failure of an item in the sample to comply with any or all requirements.

4.2.2.3 Loaded flare assemblies

4.2.2.3.1 Submission.-The five (5) loaded flare assemblies (see 4.2.1 c) shall be submitted to a Government proving ground designated by the contracting officer.

4.2.2.3.2 Inspection to be performed.-The loaded flares shall be subjected to test as follows:

a. Leakage (see 3.5) - 1 flare assembly
b. Transportation-Vibration (see 3.6) - 4 flare assemblies
c. Functioning (see 3.7) - 4 flare assemblies

4.2.2.3.3 Rejection.-The first article sample shall be rejected if:

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a. The flare assembly fails to meet the leakage requirement specified in 3.5.

b. Any launcher platform assembly or flare assembly fails to meet the transportation-vibration requirement specified in 3.6.

c. The projectile package assembly fails to meet the requirements of 3.7 as classified in 4.3.3.7, Table II. The Government reserves the right to terminate its testing upon encountering any failure of an item in the sample to comply with any or all requirements.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.1.1 Process specification.-Prior to the start of normal production the contractor shall obtain approval from the procuring activity for the process specification, submitted with the initial production sample, which shall specify the time, temperature and pressure ranges which shall be maintained during production of the molded or extruded parts and assemblies. No deviation from these limits shall be allowed without approval from the procuring activity. At unscheduled intervals the Government shall observe the molding and extruding processes to ascertain that the approved procedures are being followed; and if not being followed, all output since the last prior approved inspection may be rejected. In the event the contractor proposes a change in the process, the proposed modification must be submitted with a new initial production lot and approved before it can be used in production.

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4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Stake (see dwg. 9229107 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical: None defined			
Major: None defined			
Minor:	AQL 1.50 percent		
201.	Length, min.....	Gage	01001
202.	Diameter, max.....	Gage	01002
203.	Hole(s) missing.....	Visual	01003
204.	Point missing.....	Visual	01004
205.	Evidence of poor workmanship (see 3.8).....	Visual	01005

4.3.2.2 Angle, end (see dwg. 92219125 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			

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Minor:	AQL 2.50 percent		
201.	Total length, min.....	Gage	02001
202.	Length across each leg, min.....	Gage	02002
203.	Wall thickness, min.....	Gage	02003
204.	Radius missing (alternate only).....	Visual	02004
205.	Evidence of poor workmanship (see 3.8).....	Visual	02005

4.3.2.3 Launcher frame (see dwg. 9229110 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 2.50 percent		
201.	Inside width, min.....	Gage	03001
202.	Diameter of large holes, min.....	Gage	03002
203.	Diameter of small holes, min.....	Gage	03003
204.	Any hole missing.....	Visual	03004
205.	Evidence of poor workmanship (see 3.8).....	Visual	03005

4.3.2.4 Hinge and leg bracket (see dwg. 9229114 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 2.50 percent		
201.	Length, min.....	Gage	04001
202.	Width, min.....	Gage	04002
203.	Height, min.....	Gage	04003
204.	Location of lip.....	Gage	04004
205.	Hole missing.....	Visual	04005
206.	Evidence of poor workmanship (see 3.8).....	Visual	04006

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4.3.2.5 Leg (see dwg. 9229113 covering a detail of deg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 2.50 percent		
201.	Inside widthm min.....	Gage	05001
202.	Wall thickness, min.....	Gage	05002
203.	Total length, min.....	Gage	05003
204.	Rib missing.....	Visual	05004
205.	Point missing.....	Visual	05005
206.	Evidence of poor workmanship (see 3.8).....	Visual	05006

4.3.2.6 Leg assembly (see dwg. 9229112 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 1.50 percent		
201.	Overall width, max.....	Gage	06001
202.	Protective coating missing.....	Visual	06002
203.	Evidence of poor workmanship (see 3.8).....	Visual	06003

4.3.2.7 Launcher assembly (see dwg. 9229108 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Launch angle label missing or improperly located.....	Visual	07001
102.	Any component missing or improperly assembled.....	Visual	07002
103.	Pointer does not move freely.....	Manual	07003

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Minor:	AQL 1.00 percent		
201.	Protective coating missing.....	Visual	07004
202.	Evidence of poor workmanship (see 3.8).....	Visual	07005

4.3.2.8 Launcher platform assembly (see dwg. 9229072 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Any component missing or improperly assembled.....	Visual	08001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.8).....	Visual	08002

4.3.2.9 Retainer, tie rod (see dwg. 9229084 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Minor diameter, max.....	Gage	09001
Minor:	AQL 1.50 percent		
201.	Length, min.....	Gage	09002
202.	Diameter.....	Gage	09003
203.	Evidence of poor workmanship (see 3.8).....	Visual	09004

4.3.2.10 Tie rod (see dwg. 9229083 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major: None defined

Minor:		AQL 1.50 percent	
201.	Total length, min.....	Gage	10001
202.	Length of thread, min (2 places)....	Gage	10002
203.	Diameter under thread.....	Gage	10003
204.	Evidence of poor workmanship (see 3.8).....	Visual	10004

4.3.2.11 Tube, launch (see dwg. 9229124 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Straightness.....	Gage	11001
102.	Concentricity of outside diameter...	Gage	11002
Minor:	AQL 1.50 percent		
201.	Inside diameter, max.....	Gage	11003
202.	Outside diameter, min.....	Gage	11004
203.	Length, min.....	Gage	11005
204.	Evidence of poor workmanship (see 3.8).....	Visual	11006

4.3.2.12 Spacer (see dwg. 9229127 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 1.50 percent		
201.	Length, min.....	Gage	12001
202.	Inside diameter.....	Gage	12002
203.	Wall thickness.....	Gage	12003
204.	Evidence of poor workmanship (see 3.8).....	Visual	12004

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4.3.2.13 Plate, breech (machined) (see dwg. 9229086 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Required areas not sandblasted.....	Visual	13001
Minor:	AQL 1.50 percent		
201.	True position of fuse hole(s) and retaining rod holes with sides.....	Gage	13002
202.	Diameter of fuse hole(s).....	Gage	13003
203.	Diameter of retaining rod hole(s)...	Gage	13004
204.	Evidence of poor workmanship (see 3.8).....	Visual	13005

4.3.2.14 End cap, bottom (drilled) (see dwg. 9229080 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Required areas not sandblasted.....	Visual	14001
Minor:	AQL 2.50 percent		
201.	True position of small hole(s).....	Gage	14002
202.	True position of large hole.....	Gage	14003
203.	Diameter of small hole(s) max.....	Gage	14004
204.	Diameter of large hole, max.....	Gage	14005
205.	Evidence of poor workmanship (see 3.8).....	Visual	14006

4.3.2.15 Cup, powder (see dwg. 9229128 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		

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Minor:	AQL 1.00 percent		
201.	Outside diameter, min.....	Gage	15001
202.	Height, min.....	Gage	15002

4.3.2.16 Charge assembly, prior to assembling powder cup (see dwg. 9229123 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Weight of black powder charge.....	Balance	16001
Minor:	None defined		

4.3.2.17 Charge assembly (see dwg. 9229123 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Charge-missing.....	Visual	17001
102.	Cover punctured or torn.....	Visual	17002
103.	Sealing compound missing where required.....	Visual	17003
104.	Powder cover not taut.....	Visual	17004
Minor:	AQL 1.00 percent		
201.	Outside diameter, max.....	Gage	17005
202.	Evidence of poor workmanship (see 3.8).....	Visual	17006

4.3.2.18 Tube, delay (see dwg. 9229096 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 0.40 percent		
101.	Wall thickness.....	Gage	18001
Minor:	AQL 1.50 percent		
201.	Length, min.....	Gage	18002
202.	Outside diameter, max.....	Gage	18003
203.	Evidence of poor workmanship (see 3.8).....	Visual	18004

4.3.2.19 Piston tube assembly (see dwg. 9229092 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Any component missing or improperly assembled.....	Visual	19001
Minor:	AQL 1.50 percent		
201.	Inside diameter of delay tube for required length.....	Gage	19002
202.	Adhesive present in vent hole.....	Visual	19003
203.	Evidence of poor workmanship (see 3.8).....	Visual	19004

4.3.2.20 Sleeve, delay (see dwg. 9229132 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Wall thickness.....	Gage	20001
Minor:	AQL 1.50 percent		
201.	Length, min.....	Gage	20002
202.	Outside diameter.....	Gage	20003
203.	Evidence of poor workmanship (see 3.8).....	Visual	20004

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4.3.2.21 Pellet, black powder (see dwg. 9229104 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Weight	Balance	21001
Minor:	AQL 1.50 percent		
201.	Outside diameter, min	Gage	21002
202.	Inside diameter, max	Gage	21003
203.	Length, min	Gage	21004

4.3.2.22 Cup, fuse end (see dwg. 9229106 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Hole missing	Visual	22001
Minor:	AQL 1.50 percent		
201.	Length, min	Gage	22002
202.	Inside diameter, max	Gage	22003
203.	Hole diameter, max	Gage	22004
204.	Evidence of poor workmanship (see 3.8)	Visual	22005

4.3.2.23 Delay fuse assembly (see dwg. 9229091 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Black powder pellet missing	Visual	23001

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Minor:	AQL 1.50 percent		
201.	Location of end cup crimps.....	Gage	23002
202.	Outside diameter, max.....	Gage	23003
203.	Delay sleeve and end cup(s) not crimped 360 degrees.....	Visual	23004
204.	Evidence of poor workmanship (see 3.8).....	Visual	23005

4.3.2.24 Flight delay assembly (see dwg. 9229089 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Length from Surface "A" to delay fuze, max.....	Gage	24001
102.	Delay fuse missing.....	Visual	24002
103.	Packing missing.....	Visual	24003
104.	Delay fuse not crimped 360 degrees (2 places).....	Visual	24004
105.	Tube missing.....	Visual	24005
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.8).....	Visual	24006

4.3.2.25 Fin (see dwg. 9229103 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 2.50 percent		
201.	Total length, min.....	Gage	25001
202.	Height, min.....	Gage	25002
203.	Length from edge to beginning of angle, min.....	Gage	25003
204.	Thickness.....	Gage	25004
205.	Evidence of poor workmanship (see 3.8).....	Visual	25005

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4.3.2.26 Fin Assembly (see dwg. 9229101 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Fin missing or loose	Visual-Manual	26001
102.	Fins overlap when restrained	Visual-Manual	26002
103.	Leading edge of fin above flush	Visual	26003
104.	Two or more welds missing	Visual	26004
Minor:	AQL 2.50 percent		
201.	Total length, max	Gage	26005
202.	Length from face of fin mount to trailing edge of fin	Gage	26006
203.	Inside diameter, max	Gage	26007
204.	Location of fins	Gage	26008
205.	One weld missing	Visual	26009
206.	Evidence of poor workmanship (see 3.8)	Visual	26010

4.3.2.27 Case, projectile (see dwg. 9229100 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 1.50 percent		
201.	Outside diameter, min	Gage	27001
202.	Inside diameter, max	Gage	27002
203.	Length, min	Gage	27003
204.	Evidence of poor workmanship (see 3.8)	Visual	27004

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4.3.2.28 Candle assembly (see dwg. 9229094 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	First fire pellet missing.....	Visual	28001
Minor:	AQL 1.00 percent		
201.	Depth to illuminant composition, min.....	Gage	28002
202.	Evidence of poor workmanship (see 3.8).....	Visual	28003

4.3.2.29 Nose assembly, prior to applying adhesive (see dwg. 9229093 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Sleeve not crimped.....	Visual	29001
Minor:	AQL 0.65 percent		
201.	Length of leader, min.....	Gage	29002

4.3.2.30 Nose assembly (see dwg. 9229093 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 1.00 percent		
201.	Adhesive missing.....	Visual	30001
202.	Evidence of poor workmanship (see 3.8).....	Visual	30002

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4.3.2.31 Flight profile assembly (see dwg. 9229090 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Location of fin assembly.....	Gage	31001
Minor:	AQL 1.00 percent		
201.	Outside diameter over candle assembly, max.....	Gage	31002
202.	Evidence of poor workmanship (see 3.8).....	Visual	31003

4.3.2.32 Projectile assembly, prior to assembling flight delay assembly and base piston (see dwg. 9229088 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Length of leader.....	Gage	32001
102.	Fiberglass sleeve or tape missing...	Visual	32002
103.	Adhesive missing where required.....	Visual	32003
104.	Ejection piston "O" ring missing or not lubricated.....	Visual	32004
Minor: None defined			

4.3.2.33 Projectile assembly (see dwg. 9229088 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			

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Major:	AQL 0.65 percent		
101.	Leader wire frayed, broken or improperly assembled.....	Visual	33001
102.	Fin assembly damaged to extent that functioning may be impaired.....	Visual	33002
103.	Preformed packing missing.....	Visual	33003
104.	Base piston not fully seated.....	Visual	33004
Minor:	AQL 1.50 percent		
201.	Length, max.....	Gage	33005
202.	Marking misleading or unidentifiable.....	Visual	33006
203.	Evidence of poor workmanship (see 3.8).....	Visual	33007

4.3.2.34 Breech plate assembly (see dwg. 9229085 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Fusing assembled incorrectly.....	Visual	34001
102.	Tubing missing.....	Visual	34002
103.	Ignition mix missing.....	Visual	34003
104.	Eyelets not crimped 360 degrees.....	Visual	34004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.8).....	Visual	34005

4.3.2.35 Breech and launch tube assembly (see dwg. 9229079 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 0.65 percent		
101.	Length from top of launcher tube to projectile, min.....	Gage	35001
102.	Charge assembly missing or improperly assembled.....	Visual	35002
103.	Adhesive missing where required.....	Visual	35003
104.	Projectile missing or improperly assembled.....	Visual	35004
Minor:	AQL 1.00 percent		
201.	"O" rings not lubricated.....	Visual	35005
202.	Evidence of poor workmanship (see 3.8).....	Visual	35006

4.3.2.36 Cover, outer, assembly (see dwg. 9229073 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Required areas not sandblasted.....	Visual	36001
Minor:	AQL 1.50 percent		
201.	Handle missing or loose.....	Visual-Manual	36002
202.	Adhesive missing where required.....	Visual	36003
203.	Evidence of poor workmanship (see 3.8).....	Visual	36004

4.3.2.37 Projectile package assembly, prior to assembling protected end caps (see dwg. 9229071 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Retaining rods improperly torqued...	Gage	37001
102.	Projectile(s) not sealed.....	Visual-Manual	37002
103.	Incorrect number of projectiles.....	Visual	37003
104.	Launch tube closure missing.....	Visual	37004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.8).....	Visual	37005

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4.3.2.28 Projectile package assembly (see dwg. 9229071 covering a detail of dwg. 9229070).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Outer cover assembly not seated.....	Visual	38001
102.	Bottom end cap not sealed.....	Visual	38002
103.	Foam missing where required.....	Visual	38003
104.	Any component missing or improperly assembled.....	Visual	38004
105.	Adhesive missing where required.....	Visual	38005
Minor:	AQL 1.50 percent		
201.	Tape pull tab missing or inadequate.....	Visual	38006
202.	Marking misleading or unidentifiable.....	Visual	38007
203.	Evidence of poor workmanship (see 3.8).....	Visual	38008

4.3.2.39 Support, packing, ammunition (see dwg. 9241109).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Tape missing or incomplete.....	Visual	39001
102.	Support damaged to the extent that the contents are exposed or likely to become exposed in subsequent handling.....	Visual	39002
Minor:	None defined		

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4.3.2.40 Bag, sealed (see dwg. 9235794).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Seal incomplete or incorrectly located	Visual	40001
102.	Bag, cut, torn or punctured	Visual	40002
Minor:	AQL 0.65 percent		
201.	Marking misleading or unidentifiable	Visual	40003

4.3.2.41 Box, packing (see dwg. 9235794).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Board broken or split	Visual	41001
102.	Box damaged to the extent that contents are exposed or likely to be exposed in subsequent handling	Visual	41002
Minor:	AQL 1.50 percent		
201.	Contents loose when shaken	Manual	41003
202.	Seal missing, unsealed or improperly positioned	Visual	41004
203.	ICC marking incorrect, incomplete or illegible	Visual	41005

4.3.3 Testing

4.3.3.1 Moisture content (see Table I) - Major defect

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TABLE I

Material	Code No.
Paper and paper products (see 3.3.1)	42001
Black powder (see dwg. 9229123)	42002
First fire pellet (see 3.3.2)	42003
Flare composition (see 3.3.3)	42004

The contractor shall provide adequate controls to insure that the materials comply with the requirement. For verification, the contractor shall select and subject to test, one sample minimum of each material from each eight hours production. A composite sample shall not be used. If the moisture content of a sample exceeds the requirement and loading has not begun, the material represented by the sample shall be rejected. If components have been loaded with a material containing excess moisture, the remaining unloaded material and the loaded components shall be rejected. The determination shall be performed as specified in 4.4.1.

4.3.3.2 Compression strength of projectile case (see dwg. 9229100) - Major defect - Code No. 27005.

4.3.3.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria 80 projectile cases shall be selected from each lot for this test. The lot shall be rejected, if during the test, two or more projectile assemblies fail to meet the requirement specified. The test shall be performed as specified in 4.4.2.

4.3.3.2.2 After three consecutive lots have complied with the criteria of 4.3.3.2.1, a sample of 32 assemblies shall be selected from each lot for this test. The lot shall be rejected if any sample fails to meet the requirements specified.

4.3.3.3 Hydrostatic test of launch tube (see dwg. 9229124) - Major defect - Code No. 11007.

4.3.3.3.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria 80 launch tubes shall be selected from each lot for this test. The lot shall be rejected, if during the test, two or more launch tubes fail to meet the requirement specified. The test shall be performed as specified in 4.4.3.

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4.3.3.3.2 After three consecutive lots have complied with the criteria of 4.3.3.3.1, a sample of 32 assemblies shall be selected from each lot for this test. The lot shall be rejected if any samples fails to meet the requirements specified.

4.3.3.4 Candlepower and burning time (see 3.4.1 and 3.4.2) - Major defects - Code No. 28004 and 28005 respectively.-Twenty (20) candle assemblies shall be selected from each lot for test. If two (2) or more assemblies fail to comply with the requirements specified, the lot shall be rejected. The test shall be performed as specified in 4.4.4.

4.3.3.5 Weld peel test (see dwg. 9229101 and dwg. 9229108) - Major defect - Code No. 26011 and 07006 respectively.-Thirty-two (32) fin assemblies and 32 launcher assemblies shall be selected from each lot for test. If any weld fails to comply with the requirements, the lot represented by the samples shall be rejected. The test shall be performed as specified in 4.4.5

4.3.3.6 Pull test of leader crimp (see dwg. 9229088) - Major defect - Code No. 33008.-Thirty-two (32) projectile assemblies shall be selected from each lot for test. If any assembly fails to comply with the requirement specified, the lot shall be rejected. The test shall be performed as specified in 4.4.6.

4.3.3.7 Transportation-vibration (see 3.6) - Major defect - Code No. 43001.-Beginning with the first lot produced and continuing until the contract is completed, a sample of ten (10) flare assemblies completely packaged and sealed shall be selected from each lot for this test as specified in 4.4.7. Following the test, the package shall be opened and the launcher inspected for structural damage. The lot represented by this sample shall be rejected if any launcher fails to meet the requirement specified. The projectile package assemblies shall be retained for the functioning test.

4.3.3.8 Leakage (3.5) - Major defect - Code No. 38006.-This test shall be performed 100 percent in accordance with the procedure specified in 4.4.8. Any assembly that fails to comply with the requirement specified shall be classed defective and removed from the lot.

4.3.3.9 Functioning (Proving Ground).-The ten (10) projectile package assemblies retained after transportation-vibration shall be submitted for this test. The assemblies shall be observed for any evidence of failure to comply with the requirements, as classified in Table II, when tested as specified in 4.4.9. The lot represented by this sample shall be accepted or rejected in accordance with the criteria specified. The test shall be conducted in accordance with Acceptance Test Procedure No. FL-2.

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TABLE II

Defect	Classification	Acceptance Number	Rejection Number	Code No.
Premature	Critical	0	1	38009
Flare fails to ignite (M60 Igniter fails or fuse fails to ignite)	Major	1	2	38010
Projectile Assembly fails to eject	Major	2	3	38011
Total burning time (While suspended) of any group of six (6) flares less than min.	Major	2	3	38012
Deployment range less than min. or greater than max.	Major	6	7	38013
Deployment altitude less than min. or greater than max.	Major	8	9	38014
Parachute defects	Major	3	4	38015
Candle fails to ignite	Major	2	3	38016

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4.3.4 Inspection equipment.-Equipment Tabulation Number ET-9229070 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.5 Dimensional control of molded and extruded parts.-To qualify a new or reworked cavity or die for use in production, a sample of at least 3 parts from each cavity or die shall be fully inspected dimensionally. Individual cavity or die identification must be provided for each mold. As a control of each cavity or die during production, the above quantity of parts from each cavity or die shall be inspected for at least the defects listed in paragraphs 4.3.5.1 through 4.3.5.8 after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. If any defective parts are found during qualification of the cavity or die, the cavity or die producing the defective part will not be used in production. If any defective parts are found when inspection is performed for control of the cavity or die, the cavity or die producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for inspection for each seperate type of defect according to Standard MIL-STD-105, Code Letter G, AQL 1.50 percent. All cavities or dies formerly removed from production because of some fault, may after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the contracting officer to substantiate the request. Contractor designs of gages and test equipment required to perform the inspections listed herein shall be forwarded to Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-ND15 for approval prior to manufacture of equipment. The following sub-paragraphs identify those items and the minimum inspections subject to the requirements of this paragraph. A defect classification of Minor is assigned to all defects listed herein.

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4.3.5.1 Cover, outer (see dwg. 9229134).

Defects	Method of Inspection	Code No.
1. Profile of interior surface, min.	Gage	44001
2. Wall thickness	Gage	44002
3. Perpendicularity of ends	Gage	44003
4. Total length	Gage	44004

4.3.5.2 Cap, end (see dwg. 9229075).

Defects	Method of Inspection	Code No.
1. Profile of interior surface, min	Gage	45001
2. Wall thickness	Gage	45002
3. Parallelism	Gage	45003
4. Length	Gage	45004
5. Wall thickness of closed end	Gage	45005

4.3.5.3 Nose (see dwg. 9229129).

Defects	Method of Inspection	Code No.
1. Total length	Gage	46001
2. Length of step	Gage	46002
3. Wall thickness over step	Gage	46003
4. Large outside diameter	Gage	46004
5. Small outside diameter	Gage	46005
6. Diameter of hole	Gage	46006
7. Concentricity of outside diameter	Gage	46007
8. Radius missing	Visual	46008
9. Chamfer missing	Visual	46009
10. Gating present on indicated surfaces	Visual	46010

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4.3.5.4 Piston, base (see dwg. 9229099).

Defects	Method of Inspection	Code No.
1. Small outside diameter	Gage	47001
2. Large outside diameter	Gage	47002
3. Small inside diameter	Gage	47003
4. Large inside diameter	Gage	47004
5. Concentricity of large outside diameter	Gage	47005
6. Concentricity of small outside diameter	Gage	47006
7. Concentricity of small inside diameter	Gage	47007
8. Concentricity of large groove diameter	Gage	47008
9. Diameter of large groove	Gage	47009
10. Diameter of small groove	Gage	47010
11. Width of large groove	Gage	47011
12. Width of small groove	Gage	47012
13. Location of large groove	Gage	47013
14. Location of small groove	Gage	47014
15. Width of slot, min.	Gage	47015
16. Depth of slot, min.	Gage	47016
17. Length to shouder	Gage	47017
18. Total length	Gage	47018

4.3.5.5 Piston, ejection (see dwg. 9229098).

Defects	Method of Inspection	Code No.
1. Outside diameter	Gage	48001
2. Large inside diameter	Gage	48002
3. Small inside diameter	Gage	48003
4. Diameter of groove	Gage	48004
5. Location of groove	Gage	48005
6. Concentricity of outside diameter	Gage	48006
7. Concentricity of small inside diameter	Gage	48007
8. Concentricity of groove diameter	Gage	48008
9. Width of groove	Gage	48009
10. Diameter of hole	Gage	48010

4.3.5.6 Cap, end, protected (see dwg. 9229076).

Defects	Method of Inspection	Code No.
1. Length	Gage	49001
2. Width	Gage	49002
3. Location of rubber protector	Gage	49003
4. Height	Gage	49004

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4.3.5.7 Support, tube, launch (see dwg. 9229078).

Defects	Method of Inspection	Code No.
1. True position of launch tube holes and retaining rod holes with sides	Gage	50001
2. Profile of interior surface, min	Gage	50002
3. Wall thickness	Gage	50003
4. Diameter of launch tube hole(s)	Gage	50004
5. Diameter of retaining rod hole(s)	Gage	50005
6. Thickness	Gage	50006
7. Location of protruding edge	Gage	50007
8. Diameter of counterbore, max.	Gage	50008
9. Depth of counterbore, max.	Gage	50009

4.3.5.8 Plate, breech (see dwg. 9229086)

Defects	Method of Inspection	Code No.
1. True position of top cavities with sides	Gage	13006
2. Diameter of top cavities	Gage	13007
3. Depth of top cavities	Gage	13008
4. Location of bottom cavities	Gage	13009
5. Profile of interior surface, min.	Gage	13010
6. Wall thickness	Gage	13011
7. Thickness between top and bottom cavities	Gage	13012
8. Location of protruding edge from top	Gage	13013
9. Location of protruding edge from bottom	Gage	13014
10. Thickness, min.	Gage	13015
11. Angle between top face and protruding edge	Gage	13016
12. Angle between bottom face and protuding edge	Gage	13017

4.4 Test methods and procedures.

4.4.1 Moisture content

4.4.1.1 Paper and paper products.-The moisture content shall be determined in accordance with Method 102.1 of Standard MIL-STD-1234, except that a ten (10) gram sample of each paper product shall be used and dried for two (2) hours.

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4.4.1.2 Black powder.-The moisture content of the black powder shall be determined in accordance with the procedure specified in Specification MIL-P-223.

4.4.1.3 First fire pellet and flare composition.-An accurately weighed 25 ± 5 gram portion of the sample shall be rapidly transferred to an extraction flask and 250 milliliters (ml) of special solvent (methanol with a quantity of dry sodium nitrate equal to the magnesium content of the material being tested) added. The flask shall be stoppered and the contents swirled cautiously for several minutes until the material is thoroughly dispersed. The sample shall be allowed to remain in contact with the solvent for approximately two hours. The glass stopper shall be placed with the rubber stopper and the 50 ml pipette inserted in the hole of the stopper. A 50 ml aliquot of the clear supernatant liquid shall be withdrawn and immediately transferred to the standard titration vessel containing 100 ml of the special solvent which has been titrated to the preliminary end point as described in Standard MIL-STD-1234, Method 101.4. The final end point is reached in 3.5 minutes. A blank determination shall be conducted as follows: When the preliminary end point has been reached, a 50 ml portion of the special solvent shall be immediately transferred to the titration vessel and titrated to the exact end point in 3.5 minutes as specified for the sample. The water content shall be calculated as follows:

$$\text{Percent water} = F\,[(VR-S) - (V'R-S)]\,500$$

where:

F = grams of water per ml of standard water-in-methanol solution
V = ml of Karl Fischer reagent added to the sample
V'= ml of Karl Fischer reagent added to the blank
R = ml of standard water-in-methanol solution per ml of Karl Fischer reagent
S = ml of standard water-in-methanol solution for back titration of sample
S'= ml of standard water-in-methanol solution for back titration of blank
W = sample weight, grams

4.4.2 Compression strength of projectile case.-The projectile case shall be subjected to the compression strength test specified in Standard FED-STD-406. Observation shall be made for compliance with the requirement specified.

4.4.3 Hydrostatic test of launch tube.-The launch tube shall be assembled to the test fixture specified in 4.3.4 and the required pressure applied. Observation shall be made for rupture or obvious deformation following test.

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4.4.4 Candlepower and burning time.-The candle assembly shall be supported in a suitable fixture with the ignition end up and ignited with a quickmatch. The determination of the candlepower and burning time shall be made in accordance with the procedures and equipment specified in 4.3.4.

4.4.5 Weld peel test.-The welds on the fin assembly and launcher assembly shall be tested in accordance with the procedures specified in Specification MIL-W-6858.

4.4.6 Pull test of leader crimp.-One end of the leader shall be attached to the fixture and held stationary. The required load shall be applied along the longitudinal axis of the leader at a load of not more than one (1) inch per minute. The test equipment used shall be in accordance with 4.3.4.

4.4.7 Transportation-vibration.-The completely packaged flare assemblies shall be subjected to the transportation-vibration test specified in test 104, procedure II of Standard MIL-STD-331. After test, the packages shall be opened and the launcher assemblies shall be examined for structural damage. The projectile package assembly shall not be disassembled but shall be retained for the functioning test specified in 4.4.9. Equipment required for this test shall be as specified in 4.3.4.

4.4.8 Leakage of projectile package assembly.-The projectile package assembly shall be placed in the test chamber. A vacuum equivalent to a minimum of 6 inches of mercury (24 inches of mercury absolute pressure) shall be drawn on the chamber and held for a minimum of 60 seconds. The vacuum indicator shall be observed for a loss of vacuum during the test period. Test equipment shall be as specified in 4.3.4.

4.4.9 Functioning.-The projectile package assemblies subjected to the transportation-vibration test specified in 4.4.7 shall be utilized for this test. The assemblies shall be conditioned at plus 75 plus or minus five (5) degrees Fahrenheit for a min. of 16 hours. The assemblies shall be fired within five (5) minutes after removal from the conditioning temperature. The assembly shall be mounted in the fixture with the launch angle indicator at the 45 degree setting. The assembly shall be actuated and observation made for compliance with the requirements specified in 3.7. Test equipment shall be as specified in 4.3.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.-The flare assemblies shall be preserved and packaged in accordance with dwg. 9241109.

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5.2 Packing

5.2.1 Level A.-The containers shall be packed in accordance with dwg. 9235794.

5.3 Marking.-The boxes shall be marked as specified on dwg. 9235794.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. In addition, they should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category A, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND5. All other designs of inspection equipment except that used for molded parts (see 4.3.5) shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Proving Ground test summary:

Lot	Sample Size	Test	Requirements
First Article	1	Leakage	See 3.5
	4	Transportation-Vibration	See 3.6
	4	Functioning	See Table II
Regular Production	10	Transportation-Vibration	See 3.6
	10	Functioning	See Table II

6.5 Test Validity.-If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with additional samples.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1370-A-265

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 OCT 64 1426

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